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## End Semester Examination 2025

Name of the Course: B. Tech (ECE)

Semester: V

Name of the Paper: Control System

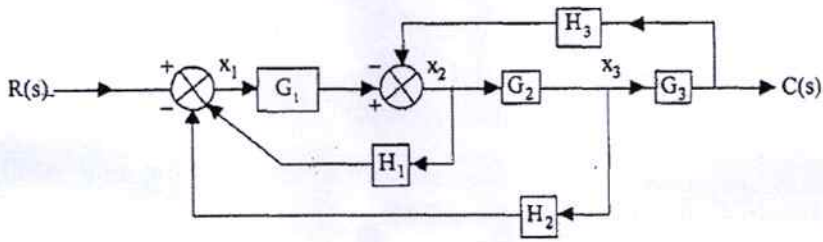
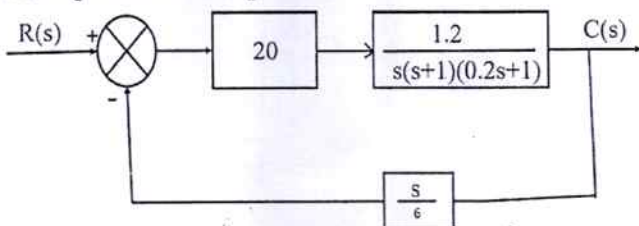
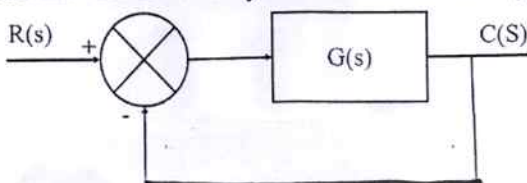
Paper Code: **TEC 541**

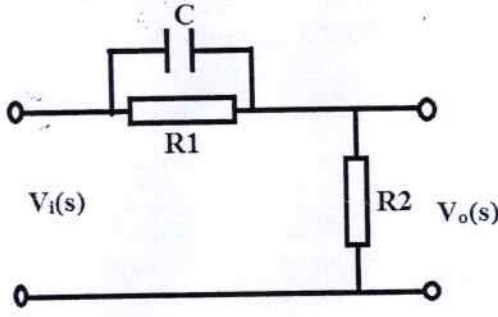
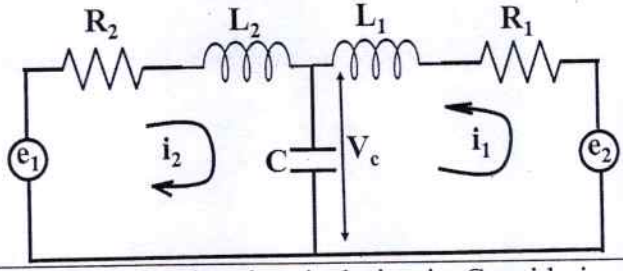
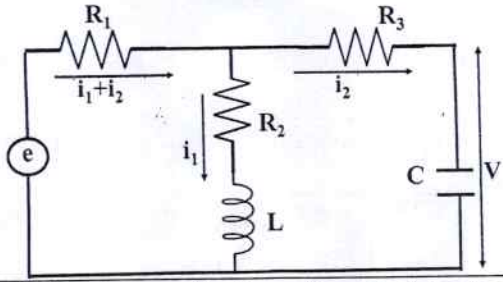
Time: 3 Hours

Maximum Marks: 100

**Note:-**

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

|           |                                                                                                                                                                                                                                                                                                 |     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Q1</b> | (20marks)                                                                                                                                                                                                                                                                                       |     |
| (a)       | Find the transfer function of the system shown in figure below using block diagram reduction technique and signal flow graph technique (Mason's Gain).                                                                                                                                          | CO1 |
|           |                                                                                                                                                                                                              |     |
| (b)       | Find the overall transfer function of electrical lead and electrical lag network.                                                                                                                                                                                                               |     |
| (c)       | Explain in brief about difference between open loop and Closed loop control Systems. Also explain their importance and stability benefits.                                                                                                                                                      |     |
| <b>Q2</b> | (20 marks)                                                                                                                                                                                                                                                                                      |     |
| (a)       | For the system shown below, determine the characteristics equation. Hence find the following when the input is unit step <div style="text-align: center;">  </div>                                          | CO2 |
|           | (i) Undamped natural frequency (ii) Damped frequency of oscillation (iii) Damping ratio and damping factor (iv) maximum overshoot (v) setting time (vi) Number of cycle completed before the output is settled within 2%, 5% of its final value.                                                |     |
| (b)       | Derive expression for time response of a second order control system subjected to unit step input function.                                                                                                                                                                                     |     |
| (c)       | The closed loop transfer function of a unity feedback control System is given below:<br>$C(s)/R(s) = (Ks+\beta)/(s^2+as+\beta)$<br>Determine steady state error for unit ramp input.                                                                                                            |     |
| <b>Q3</b> | (20 marks)                                                                                                                                                                                                                                                                                      |     |
| (a)       | In the feedback system shown in figure $G(s)=K/s(1+sT_1)(1+sT_2)$ using the Routh-Hurwitz method, determine the necessary conditions for the system to be stable. <div style="text-align: center;">  </div> | CO3 |

|           |                                                                                                                                                                                                                                                              |                   |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|           |                                                                                                                                                                                                                                                              |                   |
| (b)       | Using the Nyquist criterion investigate the closed loop stability of the system whose open loop transfer function is given below:<br>$G(s)H(s) = (s+0.25)/s^2(s+0.5)(s+1)$ .                                                                                 |                   |
| (c)       | Using Routh Hurwitz criterion investigate the stability of a unity feedback control system having following characteristics equation<br>$s^5 + s^4 + 2s^3 + 2s^2 + 11s - 10 = 0$ .                                                                           |                   |
| <b>Q4</b> | (20 marks)                                                                                                                                                                                                                                                   |                   |
| (a)       | Find the phase compensation of the following electrical network<br>                                                                                                        | CO3<br>and<br>CO4 |
| (b)       | Explain Semi-logarithmic scale and their importance in Bode Plot. Also plot the Bode plot of Gain factor K, $(1+j\omega T)$ , $1/(1+j\omega T)^N$ and $1/(j\omega T)^N$ .                                                                                    |                   |
| (c)       | Sketch the Bode plot for the open loop transfer function for the unity feedback system<br>$G(s) = 25/(s+0.5)(0.5s+1)$ .                                                                                                                                      |                   |
| <b>Q5</b> | (20 marks)                                                                                                                                                                                                                                                   |                   |
| (a)       | For the electrical network shown in fig determine the state model. Consider $i_1$ , $i_2$ , and $V_c$ as state variables. The output variables are $i_1$ and $i_2$ .<br> | CO5<br>and<br>CO6 |
| (b)       | Determine the state model for the electrical circuit. Considering $i_1$ and $V$ as state variables and $i_1+i_2$ as output.<br>                                          |                   |
| (c)       | Explain in brief about industrial application of control system.                                                                                                                                                                                             |                   |